



December 2025

ME1224-1 – TOTAL QUALITY MANAGEMENT
(For MEC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which of the following is a tool commonly used in quality control?
A) SWOT analysis
B) Fishbone diagram
C) Gantt chart
D) PESTLE analysis
- Which chart is used to monitor process variation over time?
A) Pie chart
B) Control chart
C) Bar chart
D) Scatter plot
- What does TQM primarily focus on?
A) Increasing production speed
B) Reducing employee count
C) Continuous improvement and customer satisfaction
D) Cutting marketing costs
- Which of the following is a key principle of TQM?
A) Profit maximization
B) Employee layoffs
C) Customer focus
D) Centralized decision-making
- Who is considered the father of Total Quality Management?
A) Philip Kotler
B) Peter Drucker
C) W. Edwards Deming
D) Elon Musk
- What does "Kaizen" mean in the context of TQM?
A) Large-scale innovation
B) Continuous improvement
C) Employee termination
D) Cost reduction
- Which of the following is NOT a TQM tool?
A) Control charts
B) Pareto analysis
C) Brainstorming
D) Monopoly game
- In TQM, quality is the responsibility of
A) The quality control department only
B) Top management only
C) All employees
D) External consultants
- TQM does not primarily focus on
A) Customer satisfaction
B) Employee commitment
C) Continuous improvement
D) Cutting marketing costs
- What is the center line in an $\bar{X}$ chart typically based on
A) Median of all values
B) Mode of all values
C) Average of subgroup means
D) Range of values
- Which of the following is NOT a variable control chart?
A) $\bar{X}$ chart
B) R chart
C) s chart
D) p chart
- What is the primary assumption for using variable control charts?
A) Data is binary
B) Data is qualitative
C) Data is continuous and measurable
D) Data is categorical

13. Attribute control charts are used for:
 - A) Continuous data
 - B) Variable data
 - C) Categorical data
 - D) Binary or count data
14. What does a p-chart monitor?
 - A) Mean of a process
 - B) Range of a process
 - C) Proportion of defective items
 - D) Total production
15. What does an np-chart monitor?
 - A) Number of defective items
 - B) Proportion of defects
 - C) Range of values
 - D) Standard deviation
16. What is the center line in a p-chart based on?
 - A) Average number of defects
 - B) Average proportion defective
 - C) Median of defects
 - D) Range of defects
17. What is the primary purpose of acceptance sampling?
 - A) To inspect every item
 - B) To reduce production time
 - C) To decide whether to accept or reject a lot
 - D) To improve employee performance
18. What is the Acceptable Quality Level (AQL)?
 - A) Maximum defect level a customer will tolerate
 - B) Minimum defect level in production
 - C) Average defect rate in a sample
 - D) Number of items inspected
19. What is the primary goal of Design of Experiments?
 - A) To reduce production cost
 - B) To identify cause-and-effect relationships
 - C) To increase marketing reach
 - D) To improve employee morale
20. Which of the following is NOT a type of experimental design?
 - A) Full factorial
 - B) Fractional factorial
 - C) Randomized block
 - D) Histogram

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|--|-------------|-------------|-------------|-----------|-------------|---|-------------|----|-------------|---|-------------|----|-------------|---|-------------|----|-------------|----|-------------|----|-------------|----|-------------|---|-------------|----|-------------|---|-------------|----|-------------|---|-------------|----|-------------|---|--|--|--|--|
| 1. a) Define Quality, Quality by Design and Quality Engineering. | 8 | L*2 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| b) Tests have indicated that the tensile strengths of certain aluminum castings average 1785kg/cm ² with standard deviation of 220kg/cm ² . If the distribution is normal determine the percentage of the castings that will have <ol style="list-style-type: none"> i. Tensile strength less than 1400 kg/cm² ii. Tensile strength more than 1500 kg/cm² | 8 | L2 | 2 | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 2. a) Briefly explain pareto analysis and cause and effect diagram. | 8 | L2 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| b) The weight contained in a certain product is labelled as 400 grams. Periodically, a sample is taken from the packaging line and the contents are weighed. After 250 samples, the data was recorded as follows. | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| <table border="1" style="margin: auto; border-collapse: collapse;"> <thead> <tr> <th>Weight (gm)</th> <th>Frequency</th> <th>Weight (gm)</th> <th>Frequency</th> </tr> </thead> <tbody> <tr><td>405.5-406.0</td><td style="text-align: center;">1</td><td>401.5-402.0</td><td style="text-align: center;">34</td></tr> <tr><td>405.0-405.5</td><td style="text-align: center;">2</td><td>401.0-401.5</td><td style="text-align: center;">25</td></tr> <tr><td>404.5-405.0</td><td style="text-align: center;">7</td><td>400.5-401.0</td><td style="text-align: center;">17</td></tr> <tr><td>404.0-404.5</td><td style="text-align: center;">12</td><td>400.0-400.5</td><td style="text-align: center;">13</td></tr> <tr><td>403.5-404.0</td><td style="text-align: center;">25</td><td>399.5-400.5</td><td style="text-align: center;">7</td></tr> <tr><td>403.0-403.5</td><td style="text-align: center;">31</td><td>399.0-399.5</td><td style="text-align: center;">4</td></tr> <tr><td>402.5-403.0</td><td style="text-align: center;">39</td><td>398.5-399.0</td><td style="text-align: center;">2</td></tr> <tr><td>402.0-402.5</td><td style="text-align: center;">30</td><td>398.0-398.5</td><td style="text-align: center;">1</td></tr> </tbody> </table> | Weight (gm) | Frequency | Weight (gm) | Frequency | 405.5-406.0 | 1 | 401.5-402.0 | 34 | 405.0-405.5 | 2 | 401.0-401.5 | 25 | 404.5-405.0 | 7 | 400.5-401.0 | 17 | 404.0-404.5 | 12 | 400.0-400.5 | 13 | 403.5-404.0 | 25 | 399.5-400.5 | 7 | 403.0-403.5 | 31 | 399.0-399.5 | 4 | 402.5-403.0 | 39 | 398.5-399.0 | 2 | 402.0-402.5 | 30 | 398.0-398.5 | 1 | | | | |
| Weight (gm) | Frequency | Weight (gm) | Frequency | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 405.5-406.0 | 1 | 401.5-402.0 | 34 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 405.0-405.5 | 2 | 401.0-401.5 | 25 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 404.5-405.0 | 7 | 400.5-401.0 | 17 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 404.0-404.5 | 12 | 400.0-400.5 | 13 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 403.5-404.0 | 25 | 399.5-400.5 | 7 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 403.0-403.5 | 31 | 399.0-399.5 | 4 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 402.5-403.0 | 39 | 398.5-399.0 | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 402.0-402.5 | 30 | 398.0-398.5 | 1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| c) Determine the average, median, mode and standard deviation of the distribution. | 8 | L3 | 2 | 1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 3. a) Briefly explain Deming's principles. | 8 | L2 | 1 | 1 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| b) Briefly explain the various measures of central tendency and dispersion. | 8 | L2 | 2 | 2 | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

Unit – II

4. a) The following are the $\bar{X}$ and R values of 4 subgroups of 5 readings.
 $\bar{X} = 10.2, 12.1, 10.8$ and 10.5
 $R = 1.1, 1.3, 0.9$ and 0.8
 The specification limits for the components are 10.7 ± 0.2 .
 Establish the control limits for the $\bar{X}$ and R chart. Will the product be able to meet the specifications?

8 L3 3 2

- b) Listed in the table are the number of bolts of cloth produced daily in a small textile mill and the corresponding number of imperfections found in these bolts. Use this data to draw suitable control chart. Is the process under control.

Days	Bolts of cloth produced	Number of imperfections
1	20	27
2	20	23
3	20	30
4	21	28
5	22	29
6	22	31
7	23	37
8	33	29
9	23	36
10	21	27

8 L3 4 2

5. a) The following table gives the $\bar{X}$ and R values for 20 samples of 5 readings.

Sample	$\bar{X}$	R	Sample	$\bar{X}$	R
1	34.0	4	11	38.4	4
2	31.6	2	12	34.0	14
3	30.8	3	13	35.0	4
4	33.8	5	14	33.8	7
5	31.6	2	15	31.6	5
6	33.0	5	16	33.0	7
7	28.2	13	17	32.6	3
8	33.8	19	18	31.8	9
9	37.8	6	19	35.6	6
10	35.8	4	20	33.0	4

- i) Determine the control limits for $\bar{X}$ bar and R charts.
 ii) What is the process capability?

8 L3 3 2

- b) The following table gives the number of missing rivets noted in an aircraft final inspection. Determine $\bar{C}$ control limits and state whether the process is in control. If not, then suggest the new control limits.

Aircraft No.	1	2	3	4	5	6	7	8	9
No. of missing rivets	8	16	14	19	11	15	8	11	21
Aircraft No.	10	11	12	13	14	15	16	17	18
No. of missing rivets	12	23	16	9	25	15	9	9	14
Aircraft No.	19	20	21	22	23	24	25		
No. of missing rivets	11	9	10	22	7	28	9		

8 L3 4 2

6. a) Control charts for $\bar{X}$ and R are maintained on a certain quality characteristic of a manufactured product. The subgroup size is 7. The values of $\bar{X}$ and R are computed for each subgroup. After 35 subgroups, $\sum \bar{X} = 7085$ and $\sum R = 1200$. Compute the trial control limits for the above charts. The specification on the quality characteristic is 220 ± 35 . What is the process capability? What percentage of the components is likely to be rejected?

8 L3 3 2

- b) Certain product is given 100% inspection as it is manufactured, and the resultant data is summarized by the hour. In the following table, 16 hours of data are recorded. Calculate the control limits using 3 - sigma control limits and indicate the values that are out of control. Draw the control chart.

Hour	Number of units inspected	Number of defective units	Hour	Number of units inspected	Number of defective units
1	48	5	9	32	5
2	36	5	10	40	2
3	50	0	11	47	2
4	47	5	12	47	4
5	48	0	13	46	1
6	54	3	14	46	0
7	50	0	15	48	3
8	42	1	16	39	0

8 L3 4 2

Unit – III

7. a) Define the following

- i. Producer's risk
- ii. Consumer's risk
- iii. AOQ
- iv. LTPD

8 L3 5 2

- b) Briefly explain acceptance sampling using a double sampling plan.

8 L2 5 1

8. a) Briefly explain the characteristics of the t-distribution.

8 L3 5 2

- b) A lawnmower-manufacturing company would like to advertise their lawnmower can go 250 hours before experiencing starting problems (more than two pulls) and subsequent service. During year-round field test of 10 identical lawnmowers at a proving ground in a warm climate, the 10 lawnmowers required service after 205, 430, 301, 199, 222, 124, 89, 289, 260 and 214 hours. Based on this data, can the manufacturer make this advertising claim?

8 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
